

REETO GHOSH

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EDUCATION

University of Waterloo

Bachelor of Mathematics (Honors), Statistics & Computational Mathematics

Waterloo, ON

Sep 2022 - Apr 2027

Coursework: Statistical Classification, ML in Economics, Computational Statistics & Data Analysis, Linear Models, Stochastic Processes, Data Structures & Algorithms

EXPERIENCE

Theory Ventures

AI Engineer Intern | Python, FastAPI, PostgreSQL, GCP, FastMCP

San Francisco, CA

May 2026 - Aug 2026

- Architected an agentic notification system using FastAPI, PostgreSQL, and FastMCP on Cloud Run, routing scheduled workflows to investors while executing background due diligence to autonomously surface companies and founders.
- Deployed isolated agent sandboxes to securely dispatch automated research tasks to interactive Google Chat cards.
- Instrumented eval sets using investor workflows to benchmark open models, refine tool descriptions, and optimize skills.
- Re-architected JSON pipelines to SQL for 107k+ entities, slashing deal scan runtimes from 9.5 minutes to <30s.

Wealthsimple

Software Engineer Intern | Python, SQL, dbt, Airflow, Preset

Toronto, ON

Jan 2026 - Apr 2026

- Engineered end-to-end dbt pipelines for 169k+ records, integrating mobile events to 2x measurable user adoption.
- Stabilized Airflow DAGs against bad source data, processing 4,100+ geocoded addresses with a 99.7% match score.
- Refactored SQL models to stitch legacy sources, recovering 18+ months of lost metrics and 40k monthly sessions.
- Deployed ML enrichments across 117M+ transactions, escalating a 1.3M record coverage gap via strict reconciliation.

Toronto Stock Exchange

Software Engineer Intern | Python, SQL, Presto, Hive, Apache Spark

Toronto, ON

May 2025 - Aug 2025

- Deployed a trader classification model scoring 50k+ daily participants, improving anomaly detection by 15%.
- Engineered scalable Python/SQL pipelines connecting Presto, processing 5TB+ of daily trading data for analytics.
- Migrated 500+ daily analytical queries to AWS Athena, refactoring code to cut latency by 70% and compute costs by 20%.

Bank of Canada

Software Engineer Intern | Python, PyTorch, Databricks, Azure Data Factory

Ottawa, ON

Sep 2024 - Dec 2024

- Engineered Databricks & PySpark ETL pipelines, standardizing 10M+ daily commodities records in Unity Catalog.
- Deployed a production NLP pipeline using fine-tuned BERT, scoring sentiment across 100k+ news websites & articles.
- Modeled macroeconomic data using ARIMA and GAMs to identify key variates for a national GDP estimation research paper.

Ciena Corporation

Software Engineer Intern | Python, SQL

Ottawa, ON

May 2023 - Aug 2023

- Engineered a Python pipeline processing 5GB+ of daily logs, automating Jira issue creation via keyword detection.
- Developed an interactive OpenSearch dashboard to visualize system errors, accelerating triage for 70+ engineers.
- Designed a Flask/PostgreSQL API to automate test-result storage and querying, saving 200+ hours annually.

PROJECTS

Code from Captures Research Paper | Python, TensorFlow, Transformers

- Implemented a multimodal vision-to-code pipeline translating UI screenshots to HTML/CSS and presented comparative model findings at the Canadian Undergraduate Conference on AI (CUCAI).

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript, C, Bash, Racket

Frameworks & Architecture: FastAPI, Pydantic (v2), SQLAlchemy, Alembic, Docker, FastMCP, ASGI/Uvicorn, REST APIs, JSONB

Cloud & Infrastructure: GCP (Cloud Run, Cloud Run Jobs, Cloud Scheduler, Cloud SQL, Secret Manager, WIF), PostgreSQL

Data & ML Engineering: dbt, Airflow, Databricks, PySpark, Redshift, Presto, PyTorch, Transformers, scikit-learn, MotherDuck

Agents & Developer Tooling: Sail SDK (Sailboxes), OpenCode, ContextVars, W3C Tracing, OAuth/OIDC, HMAC, pytest, uv, Ruff